Pistachio Supply Chain Optimization

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: keyword1, Keyword2, Keyword3, Keyword4

1 Introduction

The Introduction section, of referenced text [**?**] expands on the background of the work (some overlap with the Abstract is acceptable). The introduction should not include subheadings.

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2 Problem definition

The proposed network includes producers, processing centers, pistachio factories, oil extraction centers, composting centers, and customers. Pistachios are transported from orchards to processing centers, producing open-mouth pistachios, raw kernels, and waste, which are then sent to factories, oil extraction centers, and composting centers, respectively. Open-mouth pistachios are roasted, packed, and shipped to customers, while pistachio oil is delivered to oil customers and cosmetic factories. Waste is converted into compost for distribution. The model aims to minimize total costs (1), including opening, transportation, and production costs, under the following capacity and demand constraints with no allowed shortages:

$$\min \; z_1 + z_2 + z_3 \tag{1}$$

$$\text{s.t.} \quad \sum_{j=1}^{J} X_{ij} \leq Cpa_i, \quad \forall i \in I \tag{2}$$

$$\sum_{i=1}^{I} X_{ij} \leq Cpu_j \times U_j, \quad \forall j \in J \tag{3}$$

$$\sum_{e=1}^{E} Ow_{eq} + \sum_{j=1}^{J} Gw_{jq} \leq Cpy_q \times T_q, \quad \forall q \in Q \tag{4}$$

$$\sum_{j=1}^{J} Go_{jk} \leq Cpw_k \times W_k, \quad \forall k \in K \tag{5}$$

$$\sum_{j=1}^{J} Gr_{je} \leq Cpr_e \times R_e, \quad \forall e \in E \tag{6}$$

$$\sum_{e=1}^{E} O_{en_2} \leq Cpv_s \times V_s, \quad \forall s \in S \tag{7}$$

$$\sum_{k=1}^{K} Go_{jk} + \sum_{e=1} Gr_{je} + \sum_{q=1}^{Q} Gw_{jq} \leq \sum_{i=1} X_{ij}, \quad \forall j \in J \tag{8}$$

$$\sum_{k=1}^{K} Go_{jk} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_1, \quad \forall j \in J \tag{9}$$

$$\sum_{e=1}^{K} Gr_{je} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_2, \quad \forall j \in J \tag{10}$$

$$\sum_{q=1}^{Q} Gw_{jq} = \sum_{i=1}^{I} X_{ij}\theta_3, \quad \forall j \in J \tag{11}$$

$$\theta_1 + \theta_2 + \theta_3 = 1 \tag{12}$$

$$\sum_{n_1=1}^{N_1} P_{kn_1} \leq \sum_{j=1}^{J} Go_{jk} \times \gamma k, \quad \forall k \in K \tag{13}$$

$$\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \leq \sum_{j=1} Gr_{je} \times (1-\lambda), \quad \forall e \in E \tag{14}$$

$$\sum_{q=1}^{Q} Ow_{eq} \leq \sum_{j=1}^{J} Gr_{je} \times \lambda k, \quad \forall e \in E \tag{15}$$

$$\sum_{n_3=1}^{N_3} L_{sn_1} \leq \sum_{e=1}^{E} Oc_{es} \times \gamma s, \quad \forall s \in S \tag{16}$$

$$\sum_{m=1}^{M} D_{qm} \leq \left(\sum_{j=1}^{J} Gw_{jq} + \sum_{e=1}^{E} Ow_{eq} \right) \times \gamma q, \quad \forall q \in Q \tag{17}$$

$$\sum_{k=1}^{K} P_{kn_1} \leq Dp_{n_1}, \quad \forall n_1 \in N_1 \tag{18}$$

$$\sum_{e=1}^{E} O_{en_2} \leq Du_{n_2}, \quad \forall n_2 \in N_2 \tag{19}$$

$$\sum_{s=1}^{S} L_{sn_3} \leq Ds_{n_3}, \quad \forall n_3 \in N_3 \tag{20}$$

$$\sum_{q=1}^{Q} D_{qm} \leq Dc_m, \quad \forall m \in M \tag{21}$$

- The objective function (1) minimizes the total costs of the supply chain, including opening, transportation, and production costs. The costs are divided into three components: z_1 (22), z_2 (23), and z_3 (24), which represent the costs of opening facilities, transportation, and production, respectively as shown in the following equations:

$$z_1 = \sum_{j=1}^{J} Fu_j \times U_j + \sum_{q=1}^{Q} Fy_q \times Y_q + \sum_{k=1}^{K} Fw_k \times W_k + \sum_{e=1}^{E} Fr_e \times R_e + \sum_{s=1}^{S} Fv_s \times V_s \tag{22}$$

$$\begin{aligned} z_2 = & \sum_{i=1}^{I} \sum_{j=1}^{J} CI_i \times X_{ij} + \sum_{j=1}^{J} \sum_{k=1}^{K} Cu_{1j} \times Go_{jk} + \sum_{j=1}^{J} \sum_{e=1}^{E} Cu_{2j} \times Gr_{je} \\ & + \sum_{q=1}^{Q} \sum_{m=1}^{M} Cy_q \times D_{qm} + \sum_{k=1}^{K} \sum_{n_1=1}^{N_1} Cw_k \times P_{kn_1} \\ & + \sum_{e=1}^{E} Cr_e \left(\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \right) + \sum_{s=1}^{S} \sum_{n_3=1}^{N_3} Cv_s \times L_{sn_3} \end{aligned} \tag{23}$$

$$
\begin{aligned}
z_3 = &\sum_{i=1}^{I}\sum_{j=1}^{J} CX_{ij} \times X_{ij} + \sum_{j=1}^{J}\sum_{k=1}^{K} CK_{jk} \times Go_{jk} + \sum_{j=1}^{J}\sum_{q=1}^{Q} CJ_{jq} \times Gw_{jq} \\
&+ \sum_{e=1}^{E}\sum_{s=1}^{S} CS_{es} \times Oc_{es} + \sum_{e=1}^{E}\sum_{n_2=1}^{N_2} CN_{en_2} \times Oc_{en_2} + \sum_{e=1}^{E}\sum_{q=1}^{Q} CQ_{eq} \times Ow_{eq} \\
&+ \sum_{s=1}^{S}\sum_{n_3=1}^{N_3} Cl_{sn_3} \times L_{sn_3} + \sum_{k=1}^{K}\sum_{n_1=1}^{N_1} Cp_{kn_1} \times P_{kn_1} + \sum_{q=1}^{Q}\sum_{m=1}^{M} Cd_{qm} \times D_{qm} \quad (24)
\end{aligned}
$$

- The variables R_j, Y_q, W_k, Z_e, and V_s are binary decision variables that indicate whether certain elements in the model are active. For example, $R_j = 1$ if facility j is open, and 0 otherwise. The other variables in the model are continuous and positive, representing real numbers greater than or equal to zero, used for modeling flows, quantities, and other measurable factors.
- The constraints (2)-(7) address the finite capacities and demands of each center within the supply chain. Specifically, constraint (2) limits the quantity of products shipped from producers to distribution centers. Constraints (3)-(7) then ensure that the quantities of products delivered to processing centers, compost centers, pistachio factories, oil extraction centers, and cosmetic factories do not exceed the capacities of these centers.
- Products sent from any center must not exceed the orders received by that center. Therefore, constraint (8) stipulates that the quantity of products shipped from a processing center must be less than or equal to the quantity delivered to other centers. Constraints (9)-(12) specify the quantities of open-mouth pistachios, raw kernels, and waste produced through the dehulling process at each processing center. Similarly, constraints (13)-(17) regulate the flows within pistachio factories, oil extraction centers, cosmetic factories, and composting centers.
- Finally, to ensure that customer demand is met, constraints (18)-(21) guarantee that the products delivered to pistachio customers, pistachio oil customers, cosmetic customers, and compost markets meet or exceed their respective demands.

3 Heuristic approach

In this section, we present the methodology employed to address the optimization problem. The proposed approach integrates priority-based encoding and the Variable Neighborhood Search (VNS) algorithm, which together form the core of our solution strategy. We begin by detailing the solution representation, which utilizes priority-based encoding to efficiently manage and rank decision variables, allowing for effective exploration of the solution space. This encoding scheme is crucial in guiding the search process and maintaining feasibility while optimizing the objective function. Following this, we delve into the VNS algorithm, a robust metaheuristic known for its flexibility and capacity to escape local optima through systematic changes in neighborhood structures. The algorithm dynamically adjusts between multiple neighborhood structures, each designed to explore different aspects of the solution landscape, thereby enhancing the search process. We will also discuss the specific configurations and parameters used, as well as the design of neighborhood operators tailored to the problem characteristics. This comprehensive approach aims to deliver a high-quality solution by leveraging the strengths of priority-based encoding and the adaptability of VNS.

3.1 Solution representation

In this work, a priority-based encoding scheme is used to represent the solution space effectively, drawing on chromosome-based representations where each gene corresponds to a node (source or depot) in the transportation problem [1]. The transportation problem is modeled as a tree structure, with nodes representing sources or depots and edges representing transportation arcs. The primary objective is to construct a transportation tree that meets all depot demands while minimizing total transportation costs.

Each chromosome encodes the priority of nodes, guiding the construction of the transportation tree by establishing the most cost-effective connections first. The decoding process starts by identifying the node with the highest priority in the chromosome and connecting it to the corresponding node with the lowest transportation cost. For instance, if Depot 1 has the highest priority, it is linked to the source that offers the lowest cost connection to Depot 1. The shipment amount is determined by the minimum of the source's capacity and the depot's demand. Once the shipment is fulfilled, the depot's priority is updated (or set to zero if its demand is satisfied), and the process continues with the next highest priority node. This sequential process of arc appending and priority updating repeats until all depots' demands are satisfied, resulting in a complete transportation tree.

The whole supply chain itself is determined step-by-step, starting from consumer demand and moving back to producers, addressing key constraints at each stage. The priority-based encoding scheme is crucial in managing the construction process, as it ensures that the most cost-effective connections are made first, leading to an optimal solution.

3.2 VNS algorithm

The Variable Neighborhood Search (VNS) algorithm is a powerful metaheuristic that combines local search with systematic changes in neighborhood structures to explore the solution space effectively. By iteratively applying different neighborhood operators, VNS can escape local optima and discover high-quality solutions through continuous diversification and intensification phases.

The algorithm begins by initializing a solution and setting the maximum neighborhood size and time limit. It then enters a loop, where it repeatedly applies the shaking and local search procedures to generate new candidate solutions. If a better solution is found, it replaces the current best solution and resets the neighborhood size. Otherwise, the algorithm increases the neighborhood size and continues the search. The process repeats until the time limit is reached, at which point the best found solution is returned. Algorithm 1 provides a detailed outline of the VNS algorithm.

Algorithm 1 Variable Neighborhood Search (VNS)

Require: Initial solution x, maximum neighborhood size $k_{\max}$, time limit $t_{\max}$
Ensure: Best found solution x^*

$x^* \Leftarrow x$
repeat
  $k \Leftarrow 1$
  repeat
    $x' \Leftarrow \text{Shake}(x, k)$ ▷ Shaking
    $x'' \Leftarrow \text{LocalSearch}(x')$ ▷ Local search
    if $f(x'') < f(x^*)$ **then**
      $x^* \Leftarrow x''$
      $k \Leftarrow 1$ ▷ Reset neighborhood size
    else
      $k \Leftarrow k + 1$
    end if
  until $k > k_{\max}$
  $t \Leftarrow \text{CpuTime}()$
until $t > t_{\max}$

3.3 Neighborhood structures

In the context of the Variable Neighborhood Search (VNS) algorithm, neighborhood operators play a crucial role in exploring the solution space by systematically altering the current solution to generate new candidate solutions. The effectiveness of VNS relies heavily on the design and application of these operators, as they determine the diversity and direction of the search process. In this study, we utilize four classic neighborhood operators: Swap, Reversion, Insertion, and Slide. Each operator perturbs the solution in a distinct manner, allowing the algorithm to escape local optima and explore different regions of the search space. By iteratively applying these operators, VNS enhances its capability to find high-quality solutions through continuous diversification and intensification phases. The neighborhood structures used are detailed below, and illustraed in Figure 1.:

- **Swap:** The swap operator exchanges the positions of two selected elements within the chromosome. In the example, the selected elements are 6 and 3. This operator is useful for making quick adjustments to the order of elements without altering the overall structure significantly.
- **Reversion:** The reversion operator reverses the order of all elements between the two selected indices. In this case, with elements 6 and 3 chosen, the operator reverses the sequence between these two points. This operation can effectively rearrange subsequences, potentially moving the solution closer to a local optimum by drastically altering a segment of the sequence.
- **Insertion:** In this operation, the unit in the second selected position is relocated immediately after the unit in the first position. Subsequently, all units that were originally between these positions are shifted accordingly to the right. This allows the

algorithm to make focused adjustments by repositioning a specific element within the sequence while maintaining the overall order of the other elements.

- **Slide:** This operator begins by selecting two units of a solution randomly, like the other three. The first unit is removed, and the units that were initially located between the two selected units are shifted toward the start position of the first unit. Finally, the initially removed unit is placed at the position of the second unit. This sliding mechanism creates a subtle yet impactful reordering of the solution elements, which can enhance the search process by generating new configurations with minimal disruption.

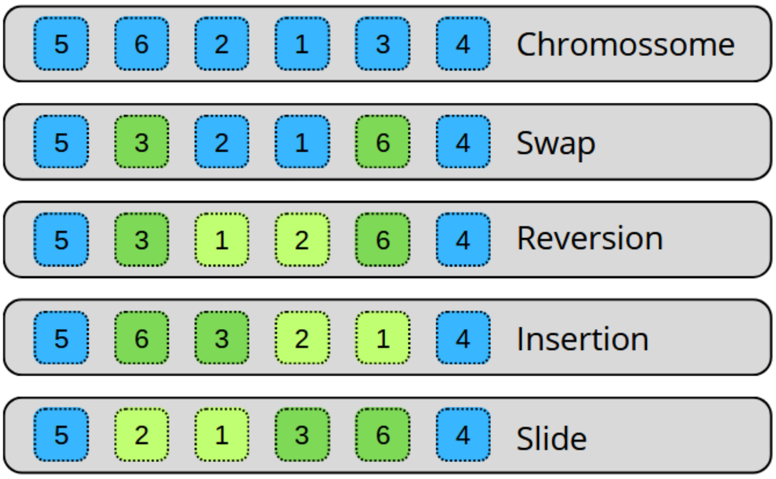

Fig. 1: Neighborhood structures used in the VNS Algorithm

4 Results

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5 Conclusion

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6 Tables

Tables can be inserted via the normal table and tabular environment. To put footnotes inside tables you should use `\footnotetext[]{...}` tag. The footnote appears just below the table itself (refer Tables 1 and 2). For the corresponding footnotemark use `\footnotemark[...]`

Table 1: Caption text

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5[1]	data 6
row 3	data 7	data 8	data 9[2]

Source: This is an example of table footnote. This is an example of table footnote.

[1]Example for a first table footnote. This is an example of table footnote.

[2]Example for a second table footnote. This is an example of table footnote.

The input format for the above table is as follows:

```
\begin{table}[<placement-specifier>]
\caption{<table-caption>}\label{<table-label>}%
\begin{tabular}{@{}llll@{}}
\toprule
Column 1 & Column 2 & Column 3 & Column 4\\
\midrule
row 1 & data 1 & data 2  & data 3 \\
row 2 & data 4 & data 5\footnotemark[1] & data 6 \\
row 3 & data 7 & data 8  & data 9\footnotemark[2]\\
\botrule
\end{tabular}
\footnotetext{Source: This is an example of table footnote.
This is an example of table footnote.}
\footnotetext[1]{Example for a first table footnote.
This is an example of table footnote.}
\footnotetext[2]{Example for a second table footnote.
This is an example of table footnote.}
\end{table}
```

Table 2: Example of a lengthy table which is set to full textwidth

	Element 1[1]			Element 2[2]		
Project	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

[1]Example for a first table footnote.

[2]Example for a second table footnote.

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*} ... \end{table*}` instead of `\begin{table} ... \end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable} ... \end{sidewaystable}` instead of `\begin{table*} ... \end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*} ... \end{sidewaystable*}`.

7 Cross referencing

Environments such as figure, table, equation and align can have a label declared via the `\label{#label}` command. For figures and table environments use the `\label{}` command inside or just below the `\caption{}` command. You can then use the `\ref{#label}` command to cross-reference them. As an example, consider the label declared for Figure **??** which is `\label{fig1}`. To cross-reference it, use the command `Figure \ref{fig1}`, for which it comes up as “Figure **??**”.

To reference line numbers in an algorithm, consider the label declared for the line number 2 of Algorithm **??** is `\label{algln2}`. To cross-reference it, use the command `\ref{algln2}` for which it comes up as line **??** of Algorithm **??**.

7.1 Details on reference citations

Standard LaTeX permits only numerical citations. To support both numerical and author-year citations this template uses `natbib` LaTeX package. For style guidance please refer to the template user manual.

Here is an example for `\cite{...}`: [**?**]. Another example for `\citep{...}`: [**?**]. For author-year citation mode, `\cite{...}` prints Jones et al. (1990) and `\citep{...}` prints (Jones et al., 1990).

All cited bib entries are printed at the end of this article: [**?**], [**?**], [**?**], [**?**], [**?**], [**?**], [**?**], [**?**], [**?**], [**?**] and [**?**].

Table 3: Tables which are too long to fit, should be written using the “sidewaystable” environment as shown here

	Element 1[1]			Element[2]		
Projectile	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40
Element 5	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 6	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

[1]This is an example of table footnote.

8 Examples for theorem like environments

For theorem like environments, we require `amsthm` package. There are three types of predefined theorem styles exists—`thmstyleone`, `thmstyletwo` and `thmstylethree`

`thmstyleone`	Numbered, theorem head in bold font and theorem text in italic style
`thmstyletwo`	Numbered, theorem head in roman font and theorem text in italic style
`thmstylethree`	Numbered, theorem head in bold font and theorem text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead)**.** *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2. *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1. *Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1. *Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Definition 1 (Definition sub head)**.** *Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.*

Additionally a predefined "proof" environment is available: `\begin{proof} ... \end{proof}`. This prints a "Proof" head in italic font style and the "body text" in roman font style with an open square at the end of each proof environment.

Proof. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proof of Theorem 1. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

For a quote environment, use `\begin{quote}...\end{quote}`

> Quoted text example. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text (refer Figure **??**). Sample body text. Sample body text. Sample body text (refer Table 3).

9 Methods

Topical subheadings are allowed. Authors must ensure that their Methods section includes adequate experimental and characterization data necessary for others in the field to reproduce their work. Authors are encouraged to include RIIDs where appropriate.

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2. Accordance: a statement explicitly saying that the methods were carried out in accordance with the relevant guidelines and regulations
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10 Discussion

Discussions should be brief and focused. In some disciplines use of Discussion or 'Conclusion' is interchangeable. It is not mandatory to use both. Some journals prefer a section 'Results and Discussion' followed by a section 'Conclusion'. Please refer to Journal-level guidance for any specific requirements.

11 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

References

[1] Gen, M., Altiparmak, F. & Lin, L. A genetic algorithm for two-stage transportation problem using priority-based encoding. *OR spectrum* **28**, 337–354 (2006).